1. b.

$$A=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$$

$$\det(A-\lambda I)=0$$

$$\det\left(\begin{bmatrix}2-\lambda & -1\\ -1 & 2-\lambda\end{bmatrix}\right)=0$$

$$(2-\lambda)(2-\lambda)-(-1)(-1)=0$$

$$\lambda^2-4\lambda+4-1=0$$

$$\lambda^2-4\lambda+3=0$$

$$(\lambda-3)(\lambda-1)=0$$

$$\lambda=3,1$$

$$\left(\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}-\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}\right)v=0$$

$$\begin{bmatrix}-1 & -1\\ -1 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=0$$

$$\begin{bmatrix}-x_1-x_2\\ -x_1-x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$x_1=-x_2$

$$v_1=\begin{bmatrix}x_1\\ -x_1\end{bmatrix}\leftarrow\begin{bmatrix}1\\ -1\end{bmatrix}$$

$\lambda_1=3$

$$\left(\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}-\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\right)v=0$$

$$\left(\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\right)\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\begin{bmatrix}x_1-x_2\\ -x_1+x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$x_1=x_2$

$$v_2=\begin{bmatrix}x_1\\ x_1\end{bmatrix}\leftarrow\begin{bmatrix}1\\ 1\end{bmatrix}$$

$\lambda_2=1$

Verifying P1.b

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2-1 \\ -1+2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \surd$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = 3\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 2+1 \\ -1-2 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix} \surd$$

P1.c Prove $\text{trace}(AB) = \text{trace}(BA)$

if ~~$AB \neq BA$~~, the traces should be equal. (Because trace is sum of diagonal)

~~$AB = BA$~~, $\text{diag}(AB) = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + \dots & 0 & 0 & \dots \\ 0 & a_{21}b_{12} + a_{22}b_{22} + \dots & 0 & \dots \\ 0 & 0 & a_{31}b_{13} + \dots & \\ \vdots & \vdots & \vdots & \end{bmatrix}$

$\text{diag}(AB) = \begin{bmatrix} a_{11}b_{11} & 0 & 0 & \dots \\ 0 & & & \\ 0 & & & \\ \vdots & & & \end{bmatrix}$

$\text{diag}(BA) = \begin{bmatrix} b_{11}a_{11} + b_{12}a_{21} + \dots & 0 & 0 & \dots \\ 0 & b_{21}a_{12} + \dots & 0 & \dots \\ 0 & 0 & b_{31}a_{13} + \dots & \dots \\ \vdots & \vdots & \vdots & \end{bmatrix}$

bad approach, ignore

Pf.

$$\text{trace}(AB) = \text{trace}(BA)$$

$$\text{trace}(AB) = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} a_{ij} b_{ji} \right)$$

$$\text{trace}(BA) = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} b_{ij} a_{ji} \right)$$

this will iterate over the same things so they are equivalent.

$$\text{trace}(M) = \sum_{i=1}^{n} m_{11} + m_{22} + \dots + m_{nn}$$

square

for $MN = R$

$$r_{ii} = \sum_{j}^{n} m_{ij} n_{ji}$$

p1.d $\begin{bmatrix} 4 & 1 \\ 2 & 6 \end{bmatrix}$

$\det(\quad) = 4 \cdot 6 - 7 \cdot 2 = 10$

$\text{inv}(\quad) = \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$ / 10

$= \begin{bmatrix} 6/10 & -7/10 \\ -2/10 & 4/10 \end{bmatrix}$

↳ non-zero, so this is invertable

p2. m Q1.a

$P(X=k) = c \cdot \frac{3^k}{k!}, \quad k = 0, 1, 2, \ldots$

$\sum_{n=0}^{\infty} c \cdot \frac{3^k}{k!} = c \cdot e^3 = 1$

$c = 1/e^3$

p2.Q1.b

$E[X] = ?$

$= \sum_{n=0}^{\infty} n \cdot \frac{3^n}{e^3 n!}$

$= \frac{1}{e^3} \sum \frac{n 3^n}{n!}$

$= \frac{1}{e^3} \cdot 3e^3$

$E[X] = 3$

$Var(X) = ?$

$= E[(X-3)^2]$

$= E[X^2 - 6X + 9]$

$= E[X^2] - 6E[X] + 9$

$= E[X^2] - E[X]^2$

$= \sum_{n=0}^{\infty} n^2 \cdot \frac{3^n}{e^3 n!} - 9$

$= \frac{1}{e^3} \sum_{n=0}^{\infty} \frac{n^2 3^n}{n!} - 9$

$= \frac{1}{e^3} \cdot 9e^3 - 9$

$= 9 - 9$

$= 0$

↳ Var(X)

P2.Q2.a

$P(A)=0.5 \quad P(B)=0.3 \quad P(A|B)=0.6$

$P(A \cap B) = P(B) \cdot P(A|B)$

$= 0.3 \cdot 0.6$

$= 0.18$

b. $P(B/A) = \frac{P(A|B)P(B)}{P(A)}$

$= 0.6 \cdot 0.3 / 0.5$

$= 0.18 / 0.5$

$= 0.36$

P2.Q2.c

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.5 + 0.3 - 0.18$

$= 0.8 - 0.18$

$= 0.62$

P2. Q3. a.

$L(p_H)/P(HHTTT) = p \cdot p \cdot p \cdot (1-p) \cdot (1-p)$

$= p^3 \cdot (p^2 - 2p + 1)$

$= p^5 - 2p^4 + p^3)$

$= p^3 \cdot (1-p)^2$

$\log(L(p_H)) = \log(p) \cdot 3 + \log(1-p) \cdot 2$

derivative $= 3/p + \frac{-2}{1-p}$

$0 = \frac{3}{p} + \frac{-2}{1-p}$

$\frac{3}{p} = \frac{2}{1-p}$

$\frac{2p}{1-p} = 3$ ↘ $2p = 3 - 3p$

$\frac{p}{1-p} = \frac{3}{2}$ $5p = 3$

$p = \frac{3-3p}{2}$ $\boxed{p = 3/5}$

P2. Q3. b.

likelihood $= \theta^3 (1-\theta)^2$

prior $= P(\theta) = \begin{cases} 2\theta, & 0 \le \theta \le 1 \\ 0, & \text{otherwise} \end{cases}$

$P(\theta | data) = \theta^3 (1-\theta)^2 \cdot 2\theta / P(data)$

$P(data) = \int_0^1 P(data|\theta) \cdot P(\theta) \, d\theta$

$= \int_0^1 \theta^3 (1-\theta)^2 \, 2\theta \, d\theta$

$= \int_0^1 2\theta^4 (\theta^2 - 2\theta + 1) \, d\theta$

$= \int_0^1 2\theta^6 - 4\theta^5 + 2\theta^4 \, d\theta$

$= \frac{2\theta^7}{7} - \frac{4\theta^6}{6} + \frac{2\theta^5}{5} \Big|_0^1$

$= \frac{2}{7} - \frac{4}{6} + \frac{2}{5} = 2/105$

$P(\theta | data) = \theta^3 (1-\theta)^2 \cdot 2\theta / 2/105$

$$\log(\theta^3(1-\theta)^2 2\theta/2/105)$$

$$\frac{105}{2}(n\log(105)+4\log(\theta)+2\log(1-\theta)$$

$$diff = \frac{4}{\theta} - \frac{2}{1-\theta}$$

$$\frac{4}{\theta} = \frac{2}{1-\theta}$$

$$2\theta = 4-4\theta$$

$$\theta = 2-2\theta$$

$$3\theta = 2$$

$$\theta = 2/3$$